Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/07/2025
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	University Park	**State:**	PA
Zip Code:	16802-6300	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/08/2012	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: New contract number is JPL 1631717 per J. Callas/JPL (Ed., 8/29/24) NOTE: End date changed to 09/30/2025 per J. Callas/JPL (Ed., 4/4/24) NOTE: End date changed to 5/31/2023 per U. Israelsson/JPL (Ed., 1/28/2020) NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 1/8/19) NOTE: End date changed to 12/31/2018 per PI (Ed., 3/10/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1631717 ; JPL 1459238		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for at least 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We will assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The initial working uncertainty budget projected an accuracy of 2.2×10^(-16). The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of others, with a goal of reaching 1×10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy. Work will also include research on space optical clocks, including the PI participating as a member of NASA's science definition team (SDT) for optical clocks and time transfer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are used in new clocks at the British, Canadian, South Korean, Polish, and Indian National Laboratories, and at CERN. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is now being used as a correction of International Atomic Time, including the correction for the PHARAO clock's frequency. More broadly, our work contributes to scientific progress towards anticipated future optical frequency atomic clocks, including testing fundamental physics with state-of-the-art optical clocks in space.
Task Progress:	The launch date of the ACES mission is currently set for April 21, 2025. In late 2020, a pre-launch accuracy evaluation of the PHARAO clock was published, which completed the bulk of the Pennsylvania State University (Penn State) work expected before launch. The Penn State group continues to interact with the PHARAO and ACES teams, addressing issues that may arise, and maintaining software for PHARAO, to establish the PHARAO clock accuracy in orbit. Work this year for PHARAO includes helping to define the initial operation modes of PHARAO. These will guide the next measurements to optimize the uncertainty of the combined velocity and density extrapolation of PHARAO’s frequency, the instability of the extrapolated PHARAO frequency correction for its high stability operating point, and the instability of its systematic uncertainty. As a member of the NASA science definition team (SDT) for optical clocks and time transfer, PI Gibble presented two talks on the "Fundamental physics with an Optical Clock Orbiting in Space" (FOCOS) mission concept: at the 2024 NASA Fundamental Physics Workshop, and the 2024 ACES International Working Group Meeting. A colloquium at the Naval Post-Graduate School and seminar at the University of Amsterdam discussed PHARAO/ACES, FOCOS, and the broader work of Gibble’s group.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Abstracts for Journals and Proceedings	Gibble K, Derevianko A, Hollberg L, Newbury N, Safronova M, Sinclair L, Yu N. "FOCOS Mission Concept – Fundamental physics with an Optical Clock Orbiting in Space. " NASA Fundamental Physics Workshop, Santa Barbara, California, May 14-16, 2024. Abstracts. 2024 NASA Fundamental Physics Workshop, Santa Barbara, California, May 14-16, 2024. , May-2024
Abstracts for Journals and Proceedings	Gibble K, Derevianko A, Hollberg L, Newbury N, Safronova M, Sinclair L, Yu N. "FOCOS Mission Concept – Fundamental physics with an Optical Clock Orbiting in Space." ACES Workshop 2024 (International Working Group Meeting), Noordwijk, The Netherlands, October 23-25, 2024. Abstracts. ACES Workshop 2024 (International Working Group Meeting), Noordwijk, The Netherlands, October 23-25, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Laurent P, Esnault F, Massonnet D, Gibble K, Salomon C. "PHARAO: The final stretch…" ACES Workshop 2024 (International Working Group Meeting), Noordwijk, The Netherlands, October 23-25, 2024. Abstracts. ACES Workshop 2024 (International Working Group Meeting), Noordwijk, The Netherlands, October 23-25, 2024. , Oct-2024